

The Solow Growth Model

Notation

- ▶ Upper-case letters denote aggregate variables.
- ▶ Lower-case letters denote per-capita variables.
- ▶ Variables with a hat, $\hat{x}$, are measured per effective unit of labour.
- ▶ A dot over a variable denotes its time derivative:

$$\dot{X}_t \equiv \frac{dX_t}{dt}.$$

Let L_t denote population at date t . Population evolves according to

$$\dot{L}_t = nL_t,$$

or equivalently,

$$\frac{\dot{L}_t}{L_t} = n,$$

where n is the constant, exogenous population growth rate.

Variables and parameters

Y_t : GDP

C_t : consumption

I_t : capital investment

K_t : capital stock

s : savings rate

A_t : productivity (technology)

x : rate of technological change

w_t : wage rate

r_t : rental rate of capital

The savings rate and technological growth rate are constant and exogenous:

$$\frac{\dot{A}_t}{A_t} = x.$$

The number of effective labour units is $A_t L_t$.

The production function

Aggregate output is

$$Y_t = F(A_t L_t, K_t).$$

Properties

1. **Constant returns to scale:**

$$F(A_t \lambda L_t, \lambda K_t) = \lambda F(A_t L_t, K_t) = \lambda Y_t.$$

2. **Positive but diminishing marginal products:**

$$F_L > 0, \quad F_K > 0, \quad F_{LL} < 0, \quad F_{KK} < 0.$$

3. **Inada conditions:** marginal products approach infinity as an input approaches zero and approach zero as an input becomes arbitrarily large.

Inada conditions - details

For labour:

$$\lim_{L_t \rightarrow 0} F_L(A_t L_t, K_t) = \infty, \quad \lim_{L_t \rightarrow \infty} F_L(A_t L_t, K_t) = 0.$$

For capital:

$$\lim_{K_t \rightarrow 0} F_K(A_t L_t, K_t) = \infty, \quad \lim_{K_t \rightarrow \infty} F_K(A_t L_t, K_t) = 0.$$

Interpretation

As an input becomes scarce, its marginal product becomes arbitrarily large. As an input becomes abundant, its marginal product approaches zero.

Cobb–Douglas production

The Cobb–Douglas production function satisfies these properties:

$$Y_t = (A_t L_t)^{1-\alpha} K_t^\alpha.$$

If factors are paid the value of their marginal products,

$$w_t = (1 - \alpha) A_t^{1-\alpha} L_t^{-\alpha} K_t^\alpha = (1 - \alpha) \frac{Y_t}{L_t},$$

and

$$r_t = \alpha (A_t L_t)^{1-\alpha} K_t^{\alpha-1} = \alpha \frac{Y_t}{K_t}.$$

Factor income shares are constant

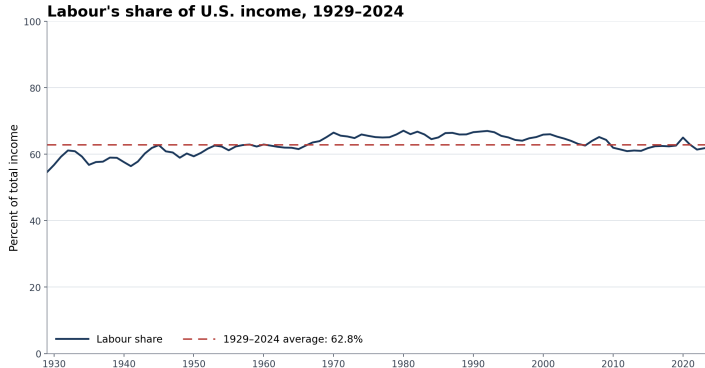
Using the factor-price equations,

$$\frac{w_t L_t}{Y_t} = 1 - \alpha \quad \text{and} \quad \frac{r_t K_t}{Y_t} = \alpha.$$

Implication

Labour's share of income is $1 - \alpha$ and capital's share is α . Thus the model has constant factor shares, a property also observed approximately in the data.

A key balanced-growth fact



A useful approximation

Labour's share fluctuates around a long-run average of 62.8% for entire period, or 64.2% for post 1960 period. Treating it as approximately constant is useful for macro modelling. A commonly used value is $1 - \alpha = 2/3 \Rightarrow \alpha = 1/3$.

Notes: Labour compensation divided by total income. Annual U.S. data, 1929–2024.

Equations of the model: aggregate and per capita

Aggregate

$$Y_t = (A_t L_t)^{1-\alpha} K_t^\alpha,$$

$$Y_t = C_t + I_t,$$

$$I_t = sY_t,$$

$$\dot{K}_t = I_t - \delta K_t.$$

Per capita

$$y_t = A_t^{1-\alpha} k_t^\alpha,$$

$$y_t = c_t + i_t,$$

$$i_t = sy_t,$$

$$\dot{k}_t = i_t - (\delta + n)k_t.$$

Equations per effective unit of labour

Define

$$\hat{y}_t = \frac{Y_t}{A_t L_t}, \quad \hat{k}_t = \frac{K_t}{A_t L_t}, \quad \hat{c}_t = \frac{C_t}{A_t L_t}, \quad \hat{i}_t = \frac{I_t}{A_t L_t}.$$

Then

$$\hat{y}_t = \hat{k}_t^\alpha, \tag{1}$$

$$\hat{y}_t = \hat{c}_t + \hat{i}_t, \tag{2}$$

$$\hat{i}_t = s\hat{y}_t, \tag{3}$$

$$\dot{\hat{k}}_t = \hat{i}_t - (\delta + n + x)\hat{k}_t, \tag{4}$$

$$r_t = \alpha \hat{k}_t^{\alpha-1}, \tag{5}$$

$$w_t = (1 - \alpha)A_t \hat{k}_t^\alpha. \tag{6}$$

Consumption and capital accumulation

From equations (2) and (3),

$$\hat{c}_t = (1 - s)\hat{y}_t. \quad (7)$$

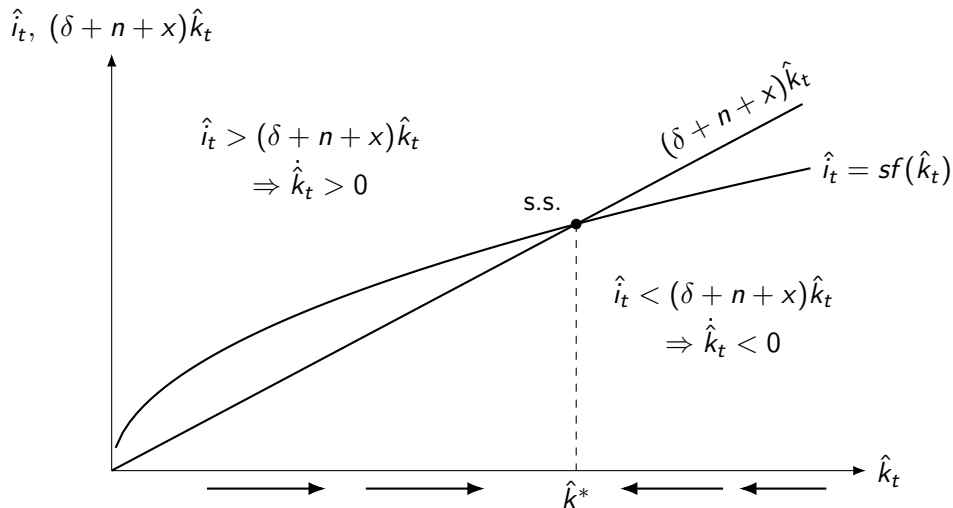
From equations (1), (3), and (4),

$$\dot{\hat{k}}_t = s\hat{y}_t - (\delta + n + x)\hat{k}_t = s\hat{k}_t^\alpha - (\delta + n + x)\hat{k}_t. \quad (8)$$

Key dynamic equation

Equation (8) is a nonlinear first-order differential equation. We want a solution that relates $\hat{k}_t$ to time, the model parameters $(s, \delta, n, x, \alpha)$, and the initial value $\hat{k}_0$.

The dynamics of the Solow model



Convergence to the steady state

For all $\hat{k}_t < \hat{k}^*$,

$$\hat{i}_t > (\delta + n + x)\hat{k}_t \implies \dot{\hat{k}}_t > 0.$$

At $\hat{k}_t = \hat{k}^*$,

$$\hat{i}_t = (\delta + n + x)\hat{k}_t \implies \dot{\hat{k}}_t = 0.$$

For all $\hat{k}_t > \hat{k}^*$,

$$\hat{i}_t < (\delta + n + x)\hat{k}_t \implies \dot{\hat{k}}_t < 0.$$

Result

The system converges to the steady state regardless of its starting position. $\hat{k}_t$ grows only during the transition to $\hat{k}^*$; once the steady state is reached, growth in capital per effective worker stops.

Do the variables in the model behave like those in the data?

The model is consistent with several long-run facts:

1. Output per capita grows at a constant rate in the long run.
2. C/Y and I/Y are constant in the long run.
3. Capital grows at the constant rate x faster than population in the long run.
4. The rate of return to capital has no long-run trend.
5. The model can generate large and persistent differences in income levels and growth rates through differences in s , A , & x .
6. & 7. *The model predicts conditional, but not absolute, convergence.*

Limitations – What the baseline model can and cannot explain

1. Key components are exogenous. The savings rate, productivity, and technological change are all exogenous. Therefore, the explanation it provides is incomplete. And the model is not well suited to policy analysis.
2. The expenditure ratios C/Y and I/Y are also constants during the transitional growth period, which is counterfactual.
3. In the baseline model (presented above) the implied speed of convergence is too high, motivating extensions to the basic framework.